ITER Vacuum Vessel — Lateral Rattle Analysis

2026-05-27 · 5,000-sample Monte Carlo · scipy HiGHS LP · 3 DOF model (Δx, Δy, Δθ) · R = 8 m · 9 gravity supports

Key numbers: LP bound (nominal assembly) = **3.09 mm** · MC P95 = **2.28 mm** · MC P99 = **2.56 mm** · Rotation range = **0.375 mrad**

§1 — Physical Mechanism

The VV sits on 9 equally-spaced four-bar linkages that allow radial motion (thermal growth, assembly). At the top of each four-bar a toroidally-oriented pin limits lateral (toroidal) travel to **±1.5 mm** about the assembly position. Any lateral force causes the VV to rattle within this envelope.

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Constraint per support i:  δᵢ = −sin(φᵢ)·Δx + cos(φᵢ)·Δy + R·Δθ  ≤  1.5 − uᵢ
                           −δᵢ                                      ≤  1.5 + uᵢ
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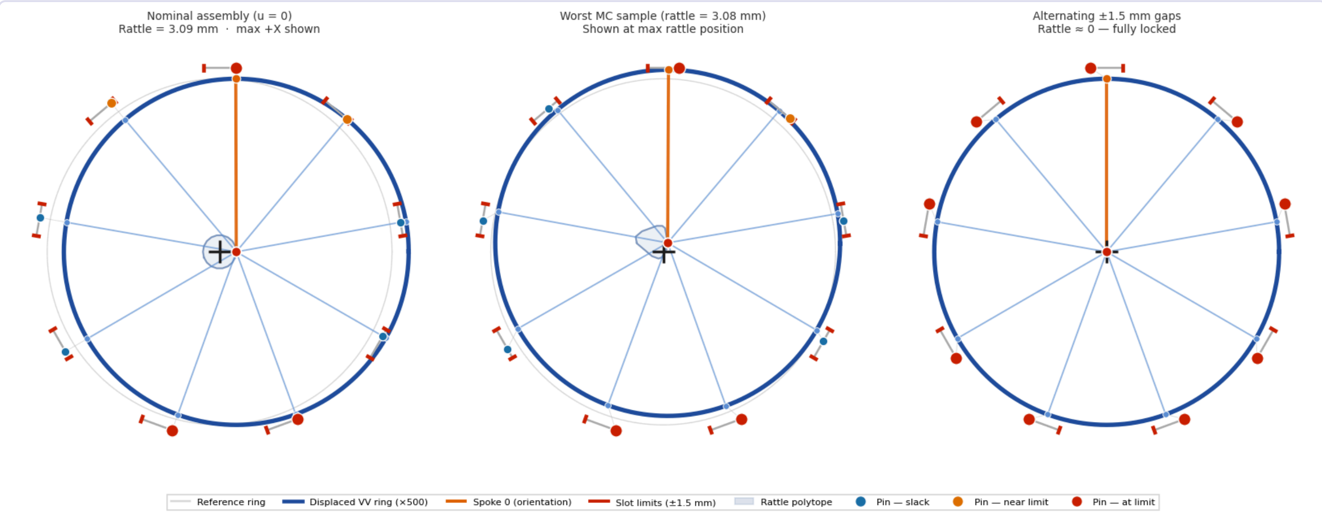
§2 — State Diagrams

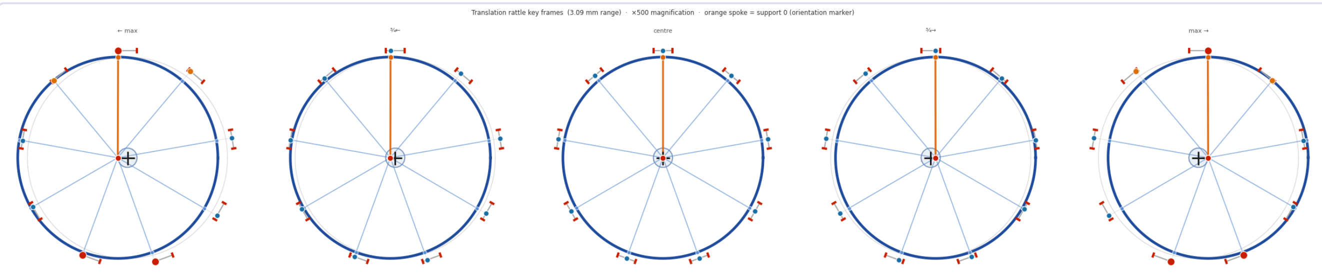
Displacements magnified ×500. Blue ring = VV position. Orange spoke = support 0 (orientation marker). Blue shaded polygon = rattle polytope (all reachable VV centre positions). Dots = pin positions.



Left: nominal (u=0), polytope ~3.09 mm diameter. Middle: worst MC sample. Right: alternating ±1.5 mm gaps — locked.

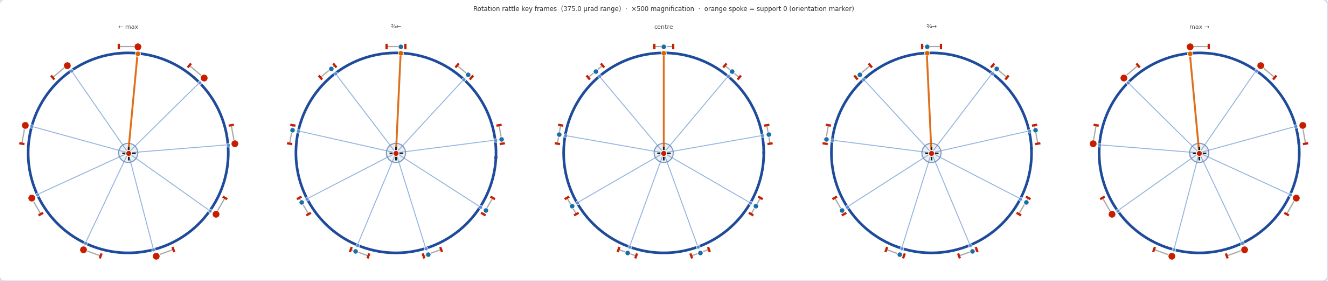
§3 — Translation Rattle Key Frames

Five key positions along the principal rattle axis. Nominal assembly (u=0), ×500 magnification.

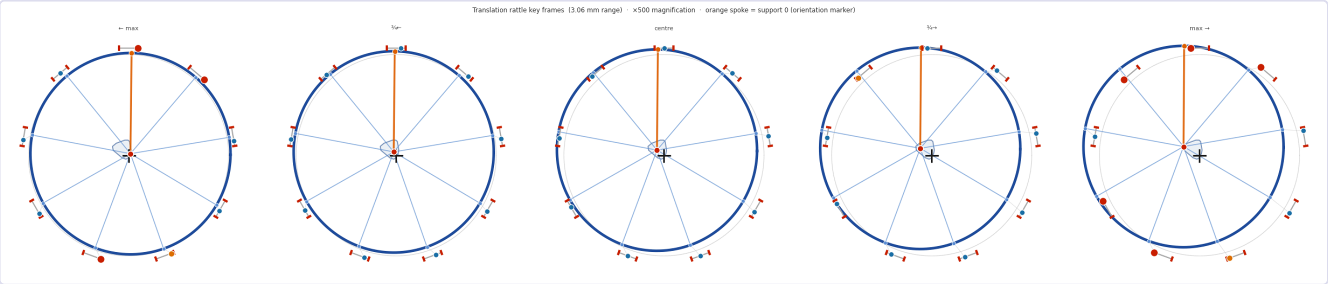


§4 — Rotation Mode Key Frames

Maximum rotation range ±187.5 µrad = ±5.37° at ×500. All 9 brackets slide synchronously.



§5 — Worst-Case Assembly: Translation Key Frames



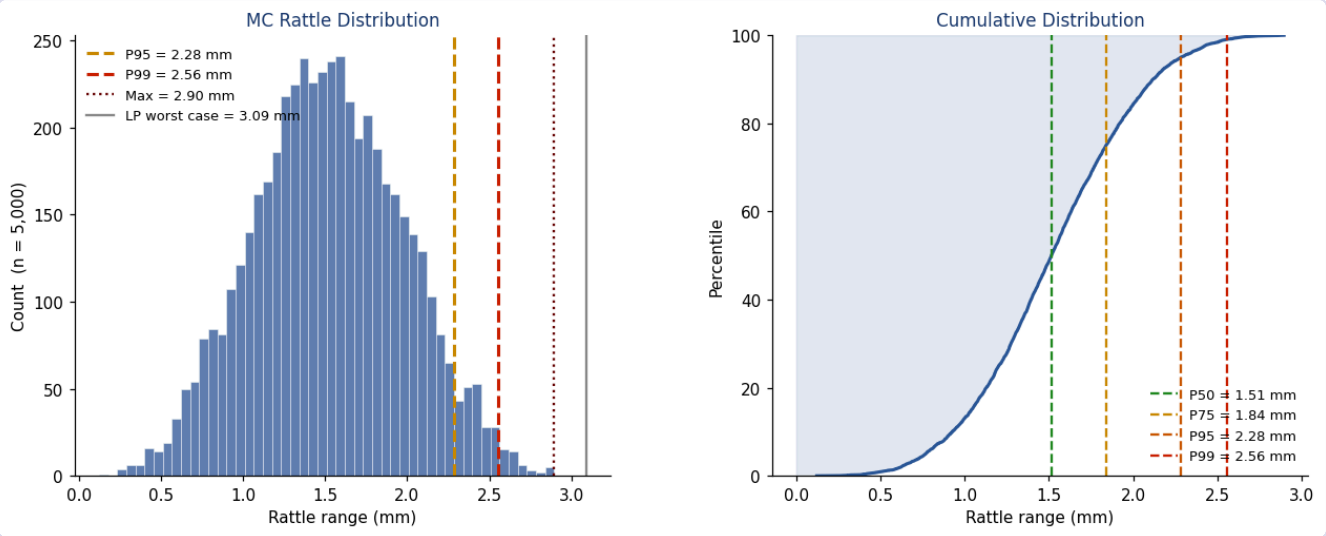
§6 — Why Rattle Exceeds 3 mm

Single bracket travel is ±1.5 mm; nominal rattle is 3.09 mm (full back-to-front range). Explanation:

Support	φ (°)	Δx coefficient	Notes
0 (top)	90	**−1.00**	LP uses rotation to neutralise this constraint
4	−70	**+0.94**	Binding at +X optimum: limits to 1.5/0.94 = 1.60 mm
5	−110	**+0.94**	Binding at +X optimum

LP simultaneously sets Δθ = +5.83 μrad, which exactly cancels the top-support constraint, allowing Δx = 1.547 mm (half-rattle). Full range = 2 × 1.547 = 3.09 mm.

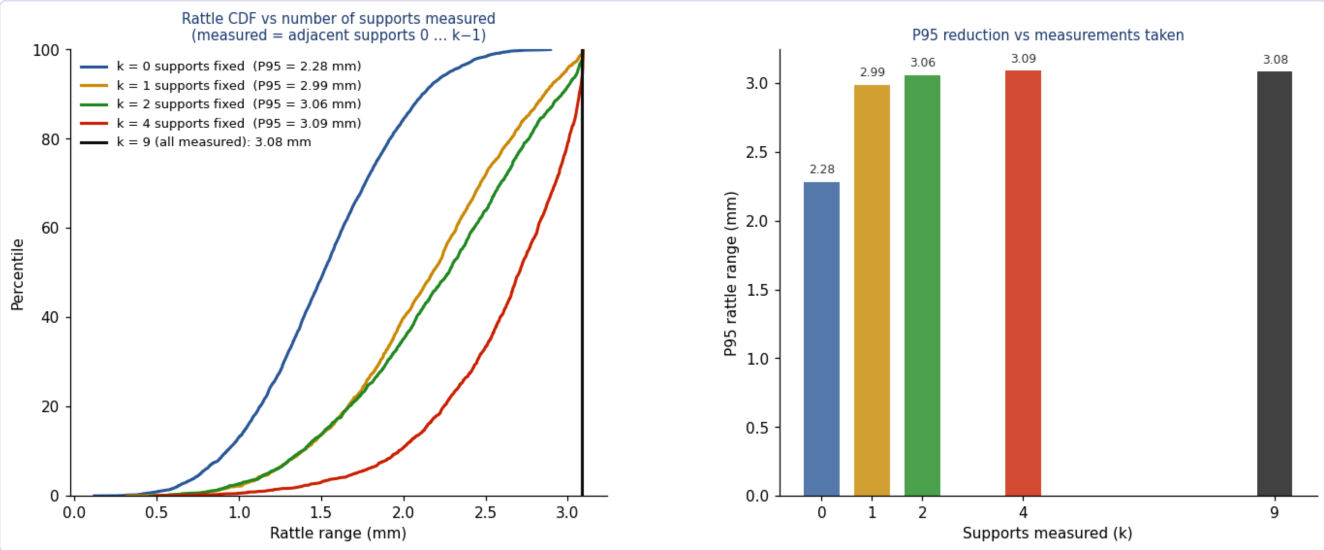
§7 — Monte Carlo Results (5,000 samples)



Rattle distribution (left) and CDF (right) for independent uniform assembly offsets.

Statistic	mm
LP bound (nominal)	**3.09**
MC P99	**2.556**
MC P95	**2.282**
MC P90	**2.119**
MC P50 (median)	**1.514**

§8 — Effect of Gap Measurements



P95 rattle vs. measured supports (left) and CDFs for 0/2/4/9 measurements (right).

> Measuring 4 adjacent supports reduces P95 from 2.28 mm to ≈1.95 mm. Measuring all 9 gives an exact deterministic answer (LP only, no MC).

§9 — Recommendations

#	Action	Status
1	Use **3.09 mm** as bounding rattle for nominal design	CONFIRMED
2	Use P95 = **2.28 mm** as statistical bound for random assemblies	CONFIRMED
3	Measure ≥4 bracket gaps before final closure	RECOMMENDED
4	Measure all 9 for exact deterministic LP answer	IDEAL
5	Include rotation (0.375 mrad) in seismic/dynamic loads analysis	RECOMMENDED
6	Verify port connections / umbilicals accommodate 3.09 mm + rotation	ACTION REQUIRED

Source: `Simon-McIntosh/vv` · `vv_rattle_mc.py` (LP solver) + `vv_viz.py` (visualisation).